NTIRE 2026 Image Denoising ($\sigma = 50$) Challenge Factsheet
Restormer with LoRA Fine-Tuning for Efficient Image Denoising

1. Team details

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- **Best scoring entry of the team during development/validation phase:** 31.94 dB
- **Link to the codes/executables of the solution:** https://github.com/your-repository

1.1. Method details

1.1.1. General Method Description

To address the Gaussian image denoising problem with noise level $\sigma = 50$, we adopt a Transformer-based image restoration framework built upon the Restormer architecture [2]. Restormer has demonstrated strong performance on various image restoration tasks due to its ability to capture long-range dependencies using self-attention mechanisms.

In our approach, we utilize a pretrained Restormer model and adapt it to the denoising task using Low-Rank Adaptation (LoRA) [1]. Instead of updating all network parameters, LoRA introduces trainable low-rank matrices into the attention projection layers while keeping the original pretrained weights frozen. This significantly reduces the number of trainable parameters and improves training efficiency while maintaining strong restoration performance.

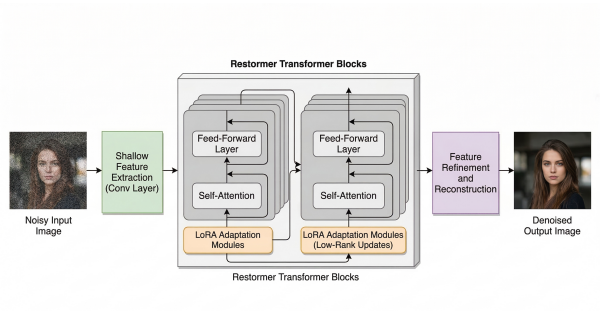

Figure 1. Overview of the proposed Restormer + LoRA architecture for Gaussian image denoising. The noisy input image is first processed through shallow feature extraction, followed by Restormer transformer blocks with LoRA-based parameter adaptation. The refined features are then reconstructed to produce the final denoised output image.

1.1.2. Model Architecture Description

The proposed architecture is built upon the Restormer encoder–decoder framework, which has demonstrated strong performance for high-resolution image restoration tasks. As illustrated in Fig. 1, the noisy input image is first processed through a shallow feature extraction module implemented using a convolutional layer to capture low-level spatial information.

The extracted shallow features are subsequently processed by a hierarchy of Restormer transformer blocks that perform deep feature extraction using multi-head self-attention and feed-forward layers. These attention mechanisms enable the network to capture both local image structures and long-range contextual dependencies, which are crucial for accurate image restoration.

To efficiently adapt the pretrained Restormer model to the denoising task, Low-Rank Adaptation (LoRA) modules are introduced into the attention projection layers of the transformer blocks. Each LoRA module introduces a pair of low-rank matrices that approximate weight updates while keeping the original pretrained parameters frozen. This parameter-efficient adaptation significantly reduces the number of trainable parameters while preserving the strong

representational capability of the backbone network.

Finally, the refined feature representations are passed through reconstruction layers that progressively map the deep features back to the RGB image space. The reconstruction module refines spatial details and produces the final denoised output image.

1.1.3. Training Details

The model is trained using a combination of the DIV2K and LSDIR datasets. To balance dataset diversity and size, a weighted sampling strategy is used during training, where approximately 12% of samples are drawn from DIV2K and 88% from LSDIR.

During training, random patches of size 256×256 are cropped from the high-resolution images. Gaussian noise with $\sigma = 50$ is added on-the-fly to generate the noisy inputs.

The network is trained for 2×10^4 iterations with a batch size of 1. The training procedure is implemented using the PyTorch framework. We adopt the Charbonnier loss function, which is widely used for image restoration tasks due to its robustness and stability.

The optimizer used is AdamW with an initial learning rate of 1×10^{-4} and weight decay of 1×10^{-4}. The learning rate is scheduled using cosine annealing to ensure stable convergence during training.

During inference, tiled processing is applied to handle high-resolution images efficiently. In addition, 8-fold test-time augmentation is employed to further improve denoising performance by averaging predictions from multiple geometric transformations of the input image.

References

[1] Edward Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, et al. Lora: Low-rank adaptation of large language models. In *ICLR*, 2022. 1

[2] Syed Waqas Zamir, Aditya Arora, Salman Khan, Munawar Hayat, Fahad Shahbaz Khan, and Ming-Hsuan Yang. Restormer: Efficient transformer for high-resolution image restoration. In *CVPR*, 2022. 1